Chromosomal microarray in practice
A published Indian case: one deletion explains the eyes, the genitals and the delay, and sets the surveillance plan

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Majjigudda RA, Menon P, Gupte S, Dikshit V, Kulkarni V, Mane S, Tamhankar PM. *A Chromosomal Microarray Detects Microdeletion at Chromosome Locus 11p14.3-p12 Leading to Wilms Tumor, Aniridia, Genitourinary Anomalies, and Mental Retardation (WAGR) Syndrome.* Cureus 2024;16(10):e72479. doi:10.7759/cureus.72479. Dr. D. Y. Patil Medical College, Hospital and Research Centre, Pune. Open access (CC BY 4.0). doi.org/10.7759/cureus.72479

The patient

- Male infant, first child of third-degree consanguineous parents; term birth, 2,100 g
- At birth: bilateral undescended testes, proximal penile hypospadias, micropenis
- 3 months: poor visual contact and light sensitivity; ophthalmology found **bilateral aniridia**
- 12 months: horizontal nystagmus; delayed motor milestones (not sitting unaided)
- Brain MRI normal; kidney ultrasound normal

Microarray finding

- Platform: CytoScan 750K
- `arr[GRCh38] 11p14.3p12(22,560,576_38,466,045)x1`
- Heterozygous deletion of about **15.9 Mb**, about 100 genes
- Includes **WT1** (Wilms tumour, genitourinary) and **PAX6** (aniridia)
- Diagnosis: **WAGR syndrome** (contiguous gene deletion)
- Parental testing not reported; no family history

How the result changed care

Because **WT1** lies in the deletion, the child was placed on **kidney ultrasound every 3 months until age 8** to screen for Wilms tumour, with kidney function monitoring from about age 10. **PAX6** loss explained the aniridia: visual stimulation, glare protection and **yearly intraocular pressure checks** for glaucoma. Surgery was planned for the testes and hypospadias, with physiotherapy and developmental therapy. The family received genetic counselling, with **prenatal microarray offered in future pregnancies**. The authors note recurrence risk is low (under 1%) when the deletion is sporadic, but higher if a parent carries a balanced rearrangement.

Talking points for the doctor visit

- CMA is the first-tier test for unexplained developmental delay/intellectual disability, autism and multiple congenital anomalies.
- A CMA report names the exact region and genes; genes such as WT1 can directly trigger a surveillance plan.
- Check parents when the CNV is uncertain or when estimating recurrence (balanced rearrangements need karyotype).
- Prenatal testing in India: written consent under the PCPNDT Act; disease testing only; fetal sex never disclosed.

For healthcare professionals' general awareness. Not a substitute for clinical judgement; facts and guidelines may change. Case details are as published by the original authors; consult the full article.

EPISODE 6 · QUICK GUIDE

Chromosomal microarray: what, when and how to read it

Topic	Key points
What it detects	Copy number variants (deletions and duplications) across the genome. G-banded karyotype resolves roughly 5-10 Mb; CMA detects much smaller changes (resolution depends on platform and lab).
When to order	First-tier: unexplained DD/ID, autism spectrum disorder, multiple congenital anomalies (ISCA consensus 2010). Prenatal: structural anomaly on ultrasound with invasive testing (ACOG Committee Opinion 682). Pregnancy loss/stillbirth tissue (no culture needed).
Yield	About 15-20% with CMA vs about 3% with karyotype in DD/ID/ASD/MCA (Miller 2010; 33 studies, 21,698 patients). India: pathogenic CNV in 18/86 (20.9%) children with intellectual disability (Boggula 2015, SGPGIMS Lucknow).
SNP array extras	Absence/runs of homozygosity (consanguinity, uniparental disomy) that can point to recessive genes; triploidy.
What it misses	Balanced translocations/inversions (karyotype); low-level mosaicism; single-gene variants (panel/exome); repeat expansions such as Fragile X (separate test).
Reading the report	`arr[GRCh38] 11p14.3p12(22,560,576_38,466,045)x1` : genome build, region, start-end, copy number (x1 = deletion, x3 = duplication). Then size, genes and class (pathogenic to benign; ACMG/ClinGen CNV standards 2020).
VUS and inherited CNVs	Test parents. Inherited from a healthy parent is not always benign (variable expressivity, e.g. 15q11.2, 16p11.2). Some CNVs may be common benign variants in Indian populations (Boggula 2015).

References

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